



Comment on “Automated detection of polymicrogyria in pediatric patients using deep learning”

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In the work presented by Guha *et al.* [1], the authors used a publicly available dataset on Kaggle [2] to develop an algorithm for the detection of polymicrogyria (PMG) in epilepsy patients. Notably, this dataset has previously been utilised by its original authors for the same detection task [3]. However, the reported near-100% accuracy raises concerns about potential methodological issues, primarily related to the understanding of the used dataset. First, the dataset is slice-labelled, yet the labels within the PMG cohort were disregarded during training and evaluation. This results in an experimental setup that diverges from the one documented in [3] and the publicly released code ¹. Second, the two cohorts, PMG and controls, differ systematically in image resolution. Consequently, the reported performance metrics likely reflect shortcut learning rather than genuine classification ability and therefore do not constitute a valid benchmark for any preprocessing pipelines or state-of-the-art methodology in PMG detection.

Below, we detail the specific misinterpretations of algorithmic performance arising from the use and handling of the dataset in the study by Guha *et al.*. Efforts for better data standards such as the Brain Imaging Data Structure (BIDS) [4] could help to alleviate such misinterpretations and avoid faulty conclusions.

The dataset comprises most, but not all, 2D slices derived from T1-weighted 3D brain magnetic resonance images (MRI) of 23 epilepsy patients with PMG, along with three matched healthy controls per epilepsy patient. Notably, slice-wise annotations are provided for the PMG cases: slices containing the malformation are labeled as positive, while the remaining slices are labeled as negative. This is clearly shown in the original manuscript and we have re-created the statistics in Figure 1(a).

Aggregating all available slices yields 4,517 2D MRI slices from PMG cases and 10,539 slices from controls, matching the class definition numbers reported by Guha *et al.*. However, when adhering to the slice-wise annotations, only 2,256 of the 4,517 slices contain PMG abnormalities. Of the remaining slices,

¹<https://github.com/RichardChangCA/Deep-Contrastive-Metric-Learning-Method-to-Detect-Polymicrogyria-in-Pediatric-Brain-MRI>

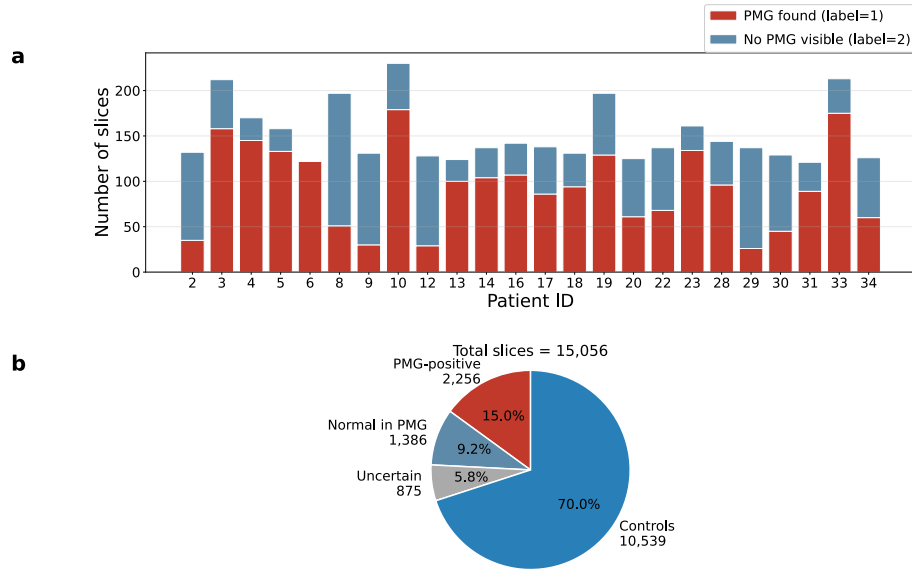


Figure 1: Overview of the 2D slices per epilepsy patient. **(a)** Actual number of labelled slices containing PMG (red) and absence of it (blue) based on the dataset published on Kaggle [2]. This histogram is a recreation of the one published by the original dataset authors in [3]. **(b)** Total proportion of available slices across PMG cases and their labels: PMG-positive, Normal in PMG and Uncertain, and controls.

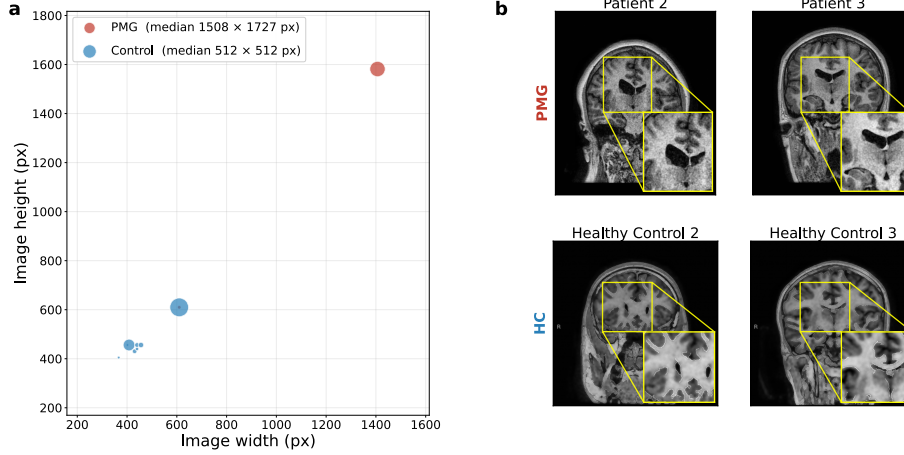


Figure 2: Overview of the 2D MRI slices resolution. **(a)** Available JPEG resolution size for each case, PMGs and controls. **(b)** Example of two different PMG cases and healthy controls, after applying the proposed preprocessing pipeline.

1,386 are considered healthy and 875 are labelled as uncertain; the latter are excluded from analysis in the original dataset publication [3]. The complete dataset statistics are shown in Figure 1(b). Importantly, including all PMG slices (regardless of annotation) not only introduces label noise but also amplifies the resolution disparities between subject groups, as illustrated in Figure 2(a).

As a result, the classification problem addressed by Guha *et al.* effectively becomes the task of detecting a resolution difference between the images. This distinction is visually apparent (see Figure 2(b)), suggesting that the task becomes a rather trivial problem. The authors further claim that the preprocessing improves the model’s ability to differentiate between cases. Due to the lack of available code, we reproduced the preprocessing pipeline based on the description provided in the Methods section. Our analysis indicates that this preprocessing step amplifies resolution differences rather than harmonising the dataset. Applying the described preprocessing to both PMG and control cases results in noticeable differences that arise simply from the resolution characteristics of the original images, as shown in Figure 2(b).

Taken together, this explains the near-100% accuracy reported by the authors when the preprocessing step is applied. However, the results are not substantially different from those obtained without such preprocessing, indicating that raw resolution differences alone are sufficient for classification. This behavior appears to be a textbook example of shortcut learning.

Guha *et al.* evaluated several state-of-the-art pretrained deep learning models, reporting DenseNet-201 as the best-performing architecture. We replicated their classification task using this same pretrained model under two conditions: (a) the slice distribution reported in their study, and (b) a corrected config-

uration reflecting the appropriate slice attribution based on the annotations. Averaged results from a 5-fold cross validation strategy are presented in Table 1. These results demonstrate that, when correct labelling is applied, performance remains high but does not reach near-100% accuracy, ~~as the PMG cases labelled as healthy introduce a degree of confounding.~~ Exact replication was not possible due to the lack of information regarding their fold distribution. However, our results under their setup (see Supplementary Table S1) are highly similar and comparable to those reported [1].

Table 1: Fine-tuning performance on a pretrained DenseNet-201 model evaluated using 5-fold cross-validation, comparing the slice distribution reported by Guha *et al.* [1] with the corrected dataset definition in [3]. Values are reported as mean $\pm$ std across folds.

Setup	Accuracy	Precision	Recall	F1	Cohen’s κ
Guha <i>et al.</i> [1]	0.980 $\pm$ 0.037	0.995 $\pm$ 0.006	0.943 $\pm$ 0.120	0.965 $\pm$ 0.068	0.951 $\pm$ 0.092
Correct	0.921 $\pm$ 0.034	0.752 $\pm$ 0.087	0.764 $\pm$ 0.130	0.755 $\pm$ 0.096	0.708 $\pm$ 0.115

In conclusion, Guha *et al.* presented a study with impressive results, claiming to ”elucidate the profound impact of image pre-processing on the classification accuracy of convolutional neural networks in the detection of Polymicrogyria”. However, these results do not reflect performance that is grounded in biological signal detection related to PMG. Instead, the preprocessing pipeline appears to exacerbate underlying resolution disparities between cohorts. While these issues originate partly from the dataset itself, they also highlight that responsible data use is as important as data sharing once the data are made available to the community [5]. Detecting PMG remains a challenging problem in medical imaging, and current state-of-the-art approaches remain limited. Consequently, publishing results that are not sufficiently supported by robust methodology does not contribute constructively to scientific progress.

Competing interests

The authors of this paper declare no competing interests.

Author contributions

The comment was conceptualized by Melanie Ganz and Llucia Coll. Data analysis was performed by Holger Lyng and Llucia Coll. Lastly, all authors contributed to writing.

References

- [1] Guha, S., Bhandage, V. & Agarwal, A. Automated detection of polymicrogyria in pediatric patients using deep learning. *Scientific Reports* **15**, 41662 (2025). URL <https://www.nature.com/articles/s41598-025-25572-6>.

- [2] Zhang, L., Abdeen, N. & Lang, J. pediatric polymicrogyria MRI dataset. URL <https://www.kaggle.com/datasets/lingfengzhang/pediatric-polymicrogyria-mri-dataset>.
- [3] Zhang, L., Abdeen, N. & Lang, J. A novel center-based deep contrastive metric learning method for the detection of polymicrogyria in pediatric brain MRI. *Computerized Medical Imaging and Graphics* **114**, 102373 (2024). URL <https://www.sciencedirect.com/science/article/pii/S0895611124000508>.
- [4] Ganz, M. & Poldrack, R. A. Data sharing in neuroimaging: experiences from the BIDS project. *Nature Reviews Neuroscience* **24**, 729–730 (2023). URL <https://www.nature.com/articles/s41583-023-00762-1>.
- [5] Gorgolewski, K. J. *et al.* The brain imaging data structure, a format for organizing and describing outputs of neuroimaging experiments. *Scientific Data* **3**, 160044 (2016). URL <https://www.nature.com/articles/sdata201644>.

Supplementary Table

Table S1: DenseNet-201 5-fold cross-validation test metrics.

(a) Paper setup (Guha & Bhandage replication)

Fold	Accuracy	Precision	Recall	F1	Cohen's κ
Fold 1	1.000	1.000	1.000	1.000	1.000
Fold 2	0.997	0.990	1.000	0.995	0.993
Fold 3	0.996	0.988	1.000	0.994	0.991
Fold 4	0.994	0.996	0.986	0.991	0.986
Fold 5	0.914	1.000	0.729	0.843	0.786

(b) Correct setup

Fold	Accuracy	Precision	Recall	F1	Cohen's κ
Fold 1	0.929	0.703	0.863	0.775	0.733
Fold 2	0.956	0.772	0.872	0.819	0.794
Fold 3	0.938	0.863	0.797	0.828	0.791
Fold 4	0.916	0.789	0.737	0.762	0.711
Fold 5	0.867	0.634	0.553	0.591	0.512